DATABASE NORMALIZATION

ASSIGNMENT FOR THE DATABASE

Lecturer :

Asmunin, S.Kom., M.Kom.

Presented by :

Aisyah Currents

25091397108 / 2025I

INFORMATIC MANAGEMENT MAJOR

FACULTY OF VOCATION

UNIVERSITAS NEGERI SURABAYA

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CASE 1

No	Nomor Mahasiswa	Nama	Hasil Ujian			
			Ke 1	Ke 2	Ke 3	Ke 4
1	01001	Ali	50	60		
2	01002	Budi	80			
3	01003	Budi	50	50	70	
4	01004	Carli	40	50		
5	01005	Dandi	59	41	42	43

- First Normal Form (1NF)

The original table contains repeating groups in the "Hasil Ujian" columns (ke-1, ke-2, ke-3, ke-4). To convert it into 1NF, these repeating columns are transformed into rows so that each cell contains only one single atomic value. The result is a table with attributes : NIM, Nama, ke-Ujian, and Nilai, where each exam attempt is stored as a seperate row.

No	NIM	Nama	Ke_Ujian	Nilai
1	01001	Ali	1	50
2	01001	Ali	2	60
3	01002	Budi	1	80
4	01003	Budi	1	50
5	01003	Budi	2	50
6	01003	Budi	3	70
7	01004	Carli	1	40
8	01004	Carli	2	50
9	01005	Dandi	1	59
10	01005	Dandi	2	41
11	01005	Dandi	3	42
12	01005	Dandi	4	43

- Second Normal Form (2NF)

After achieving 1NF, the table still has a partial dependency because the attribute "Nama" depends only on NIM and not on the composite key (NIM, Ke_Ujian). To resolve this, the table is decomposed into two tables : a. Mahasiswa, table (NIM, Nama) and a Nilai table (NIM, ke-Ujian, Nilai). This ensures that all non-key attributes are fully dependent on their respective primary keys.

- Mahasiswa table

NIM	Nama
01001	Ali
01002	Budi
01003	Budi
01004	Carli
01005	Dandi

- Nilai table

NIM	Ke-Ujian	Nilai
0001	1	50
01001	2	60
01002	1	80
01003	1	50
01003	2	50
01003	3	70
01004	1	40
01004	2	50
01005	1	59
01005	2	41
01005	3	42
01005	4	43

- Third Normal Form (3NF)

In the resulting structure, there are no transitive dependencies. The Mahasiswa table has NIM determining Nama, and the Nilai table has the composite key (NIM, Ke-Ujian) determining Nilai. Since no non-key attribute depends on another non-key attribute, the design already satisfies 3NF and does not require futher decomposition.

CASE 2

PT. SAMPLE SPECIAL
Jl. HR Rasuna SaidNo. 343
Kuningan, Setia Budi 12350, Indonesia.
Telp. : 021 5678 7768 Email : sample@special.com
Fax : 021 4564 7569 www.samplespecial.com

INVOICE
CUSTOMER - ACCOUNT

SOLD TO :
SAMPLE INDONESIA, PT.

DELIVERED TO :
SAMPLE INDONESIA, PT.
JL. IMAM BONJOL NO.60 JAKARTA PUSAT 10221
INDONESIA

Up. Mr. Dimas Prasetyo

INVOICE NO : 00000145
DATE : 11 February 2013
P/O NO : 00000234
TERMS : Cash/Tunai
PAGE NUMBER : 1

Item	Description	Qty	Unit Price	Amount
1. 1235-5677	RING	6 Pcs	19.04	114.24
2. 7655-8766	PTFE	6 Pcs	325.86	1,955.16
3. 345-9876	VALVE	1 Pcs	672.08	672.08
4. 2567-9777	BALL	12 Pcs	37.72	452.64

E & O.E.

Sub Total	3,194.12
Less : Discount	0.00
P.P.N 10%	319.41
TOTAL	0.00

IMPORTANT NOTES :
. IN CASE OF ANY DISCREPANCIES, PLEASE NOTIFY US WITHIN SEVEN (7) DAYS UPON RECEIVING OF THIS INVOICE
. INTEREST WILL BE CHARGED AT 1.5% PER MONTH ON ALL OVERDUE ACCOUNT
. NO CLAIMS WILL BE ENTERTAINED AFTER 3 WORKING DAYS FROM THE DATE OF DELIVERY
. N.E.G.- NEAREST EQUIVALENT GRADE WITHOUT PREJUDICE FOR REFERENCE ONLY
. ALL CHEQUES SHOULD BE CROSSED AND MADE PAYABLE TO
. PERMATA BANK CABANG : POLIM, JAKARTA
. IDR 7 1111 7156
. USD 7 0134 6789

for PT. SAMPLE SPECIAL

RIHANA
Authorised Signature

- First Normal Form (1NF)

In the First Stage, the Invoice data is transformed to eliminate repeating groups, Specifically the multiple item entries listed in a single invoice. Each item is converted into a seperate row so that all attributes contain atomic values, resulting in a flat table where invoice number, date, customer information, and item details are stored together. The primary key at this stage is typically a combination of Invoice_No and Item_ID, since one invoice can contain multiple items.

Invoice_No	Date	Customer_Name	Address	Item_ID	Description	Qty	Unit_price	Amount
00000 145	2013-02-11	SAMPLE INDONESIA PT	jakarta	1235-5677	RING	6 pcs	19.04	114.24
00000 145	2013-02-11	SAMPLE INDONESIA PT	jakarta	7655-8766	PTFE	6 pcs	325.86	1.955.16
00000 145	2013-02-11	SAMPLE INDONESIA PT	jakarta	345-9876	VALVE	1 pcs	672.08	672.08
00000 145	2013-02-11	SAMPLE INDONESIA PT	jakarta	2567-9777	BALL	12 pcs	37.72	452.64

- Second Normal Form (2NF)

In the Second Stage, the table is analyzed for partial dependencies. It is found that some attributes, such as Customer name and address, depend only on the invoice number, while item description and unit price depend only on the item ID. To resolves this, the table is decomposed into multiple related tables: Customer, Invoice, Item, and Invoice_Detail. This ensures that each non-key attribute is fully dependent on its respective primary key, reducing redundancy and improving data organization.

• Invoice table

Invoice_No	Date
00000 145	2013-02-11

• Customer table

Customer_Name	Address
SAMPLE INDONESIA PT	jakarta

• Item table

Item_ID	Description	Unit_Price
1235-5677	RING	19.04
7655-8766	PTFE	325.86
345-9876	VALVE	672.08
2567-9777	BALL	37.72

- Invoice detail

Invoice - No.	Item ID	Qty
00000 195	1235-5677	6
00000 195	7655-8766	6
00000 195	395-9876	1
00000 195	2567-9777	12

- Third Normal Form (3NF)
In the final stage, transitive dependencies are removed by ensuring that no non-key attribute depends on another non-key attribute. The existing structure already satisfies this condition.